

Problem 9.7

If

$$r = \frac{i^{(4)}}{d^{(4)}},$$

express v in terms of r .

Solution

Recall that

$$v = \frac{1}{1+i}.$$

For nominal interest and discount rates convertible quarterly,

$$\begin{aligned} i^{(4)} &= 4 \left((1+i)^{1/4} - 1 \right), \\ d^{(4)} &= 4 \left(1 - (1+i)^{-1/4} \right). \end{aligned}$$

Since $v = (1+i)^{-1}$, we have

$$(1+i)^{1/4} = v^{-1/4}, \quad (1+i)^{-1/4} = v^{1/4}.$$

Thus,

$$\begin{aligned} i^{(4)} &= 4 \left(v^{-1/4} - 1 \right), \\ d^{(4)} &= 4 \left(1 - v^{1/4} \right). \end{aligned}$$

Therefore,

$$\begin{aligned} r &= \frac{i^{(4)}}{d^{(4)}} \\ &= \frac{4 \left(v^{-1/4} - 1 \right)}{4 \left(1 - v^{1/4} \right)} \\ &= \frac{v^{-1/4} - 1}{1 - v^{1/4}}. \end{aligned}$$

Now simplify the numerator:

$$v^{-1/4} - 1 = \frac{1}{v^{1/4}} - 1 = \frac{1 - v^{1/4}}{v^{1/4}}.$$

So

$$\begin{aligned} r &= \frac{\frac{1-v^{1/4}}{v^{1/4}}}{1-v^{1/4}} \\ &= \frac{1}{v^{1/4}}. \end{aligned}$$

Hence,

$$r = v^{-1/4}.$$

Solving for v ,

$$\boxed{v = \frac{1}{r^4}}.$$